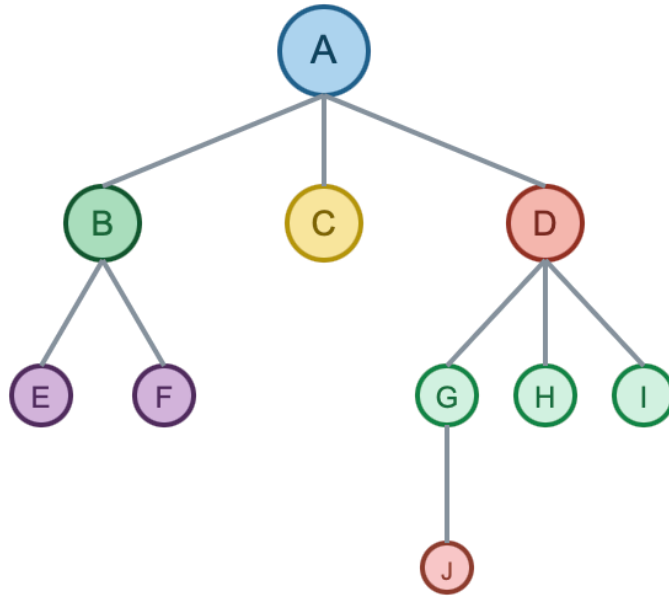


Minimum Spanning Tree



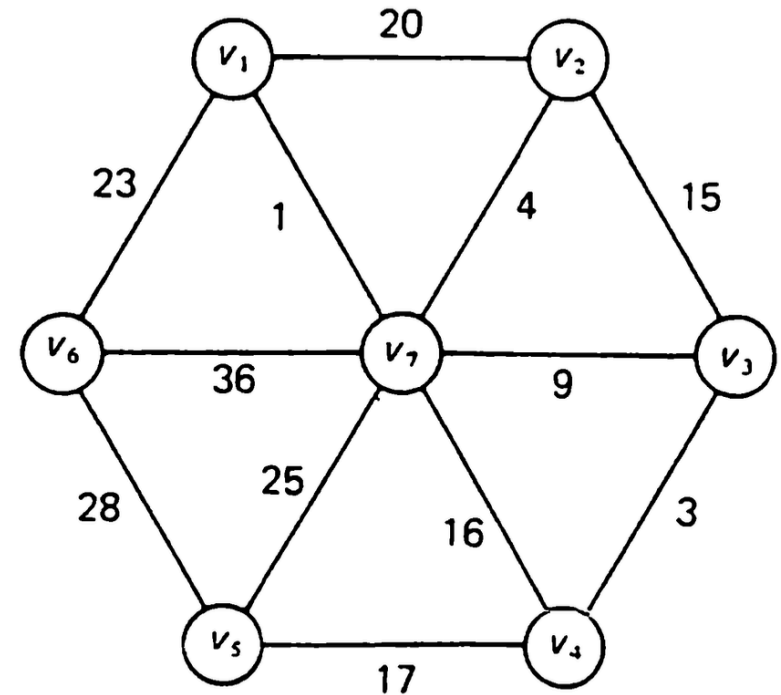
Tree



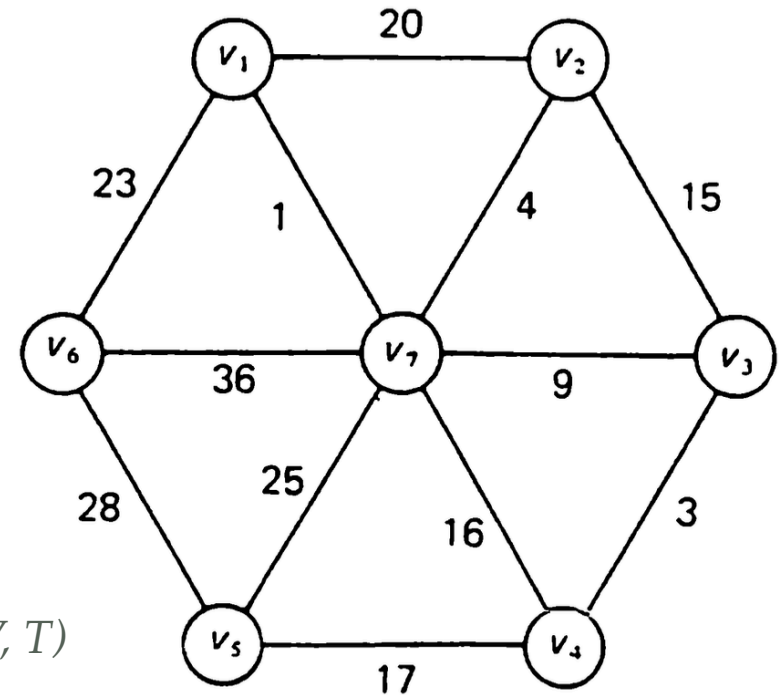
- ◆ Every pair of nodes in a tree is connected by a unique path.
- ◆ A tree does not have any cycles.
- ◆ The number of edges in a tree with n nodes is $(n - 1)$.

Minimum Spanning Tree

- ◆ Let $G = (V, E)$ be a connected, undirected graph with a cost function mapping edges to real numbers.
- ◆ A *spanning tree*, is an undirected tree that connects *all* vertices in V .
- ◆ Our goal is to find a spanning tree of minimal cost for G .

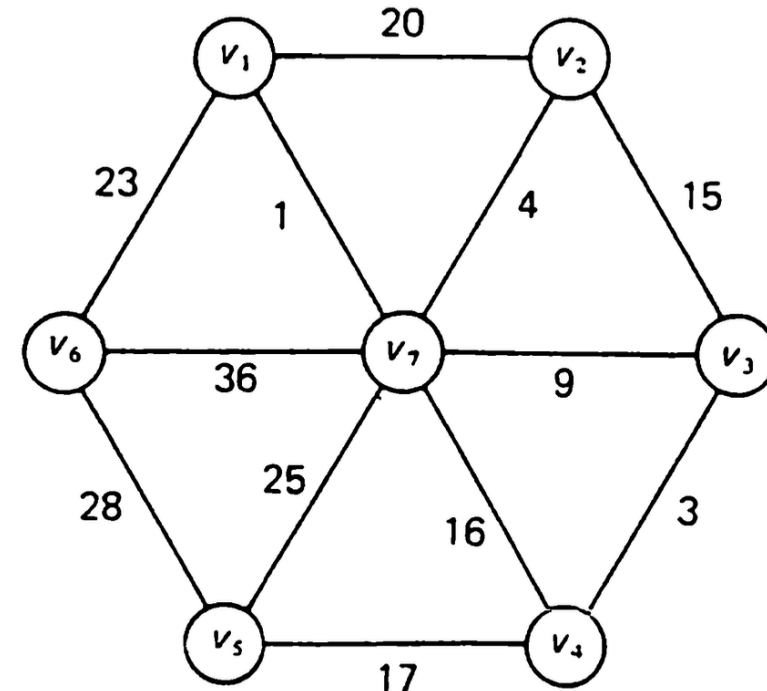


Minimum Spanning Tree



- ◆ Let $G = (V, E)$ be a connected, undirected graph and $S = (V, T)$ a spanning tree for G . Then
 - ◆ a) for all v_1 and v_2 in V , the path between v_1 and v_2 in S is unique, and
 - ◆ b) if any edge in $E - T$ is added to S , a unique cycle results.

Minimum Spanning Tree



begin

1. $T \leftarrow \emptyset$;

2. $VS \leftarrow \emptyset$;

3. construct a priority queue Q containing all edges in E ;

4. **for** each vertex $v \in V$ **do** add $\{v\}$ to VS ;

5. **while** $\|VS\| > 1$ **do**

begin

 choose (v, w) , an edge in Q of lowest cost;

 delete (v, w) from Q ;

if v and w are in different sets W_1 and W_2 in VS **then**

begin

 replace W_1 and W_2 in VS by $W_1 \cup W_2$;

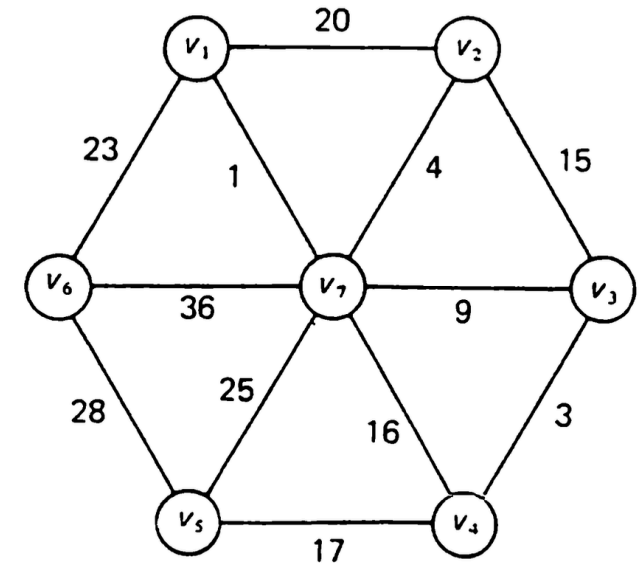
 add (v, w) to T

end

end

end

Minimum Spanning Tree



Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
(v_3, v_7)	9	(v_5, v_7)	25
(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36

begin

1. $T \leftarrow \emptyset$;

2. $VS \leftarrow \emptyset$;

3. construct a priority queue Q containing all edges in E ;

4. **for** each vertex $v \in V$ **do** add $\{v\}$ to VS ;

5. **while** $\|VS\| > 1$ **do**

begin

6. choose (v, w) , an edge in Q of lowest cost;

7. delete (v, w) from Q ;

8. **if** v and w are in different sets W_1 and W_2 in VS **then**

begin

9. replace W_1 and W_2 in VS by $W_1 \cup W_2$;

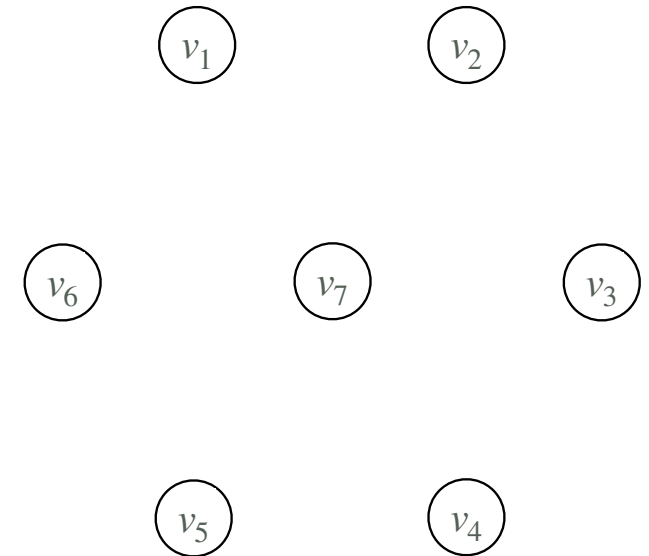
10. add (v, w) to T

end

end

end

Minimum Spanning Tree



Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
(v_3, v_7)	9	(v_5, v_7)	25
(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36

begin

1. $T \leftarrow \emptyset$;
 2. $VS \leftarrow \emptyset$;
 3. construct a priority queue Q containing all edges in E ;
 4. **for** each vertex $v \in V$ **do** add $\{v\}$ to VS ;
 5. **while** $\|VS\| > 1$ **do**

begin

 6. choose (v, w) , an edge in Q of lowest cost;
 7. delete (v, w) from Q ;
 8. **if** v and w are in different sets W_1 and W_2 in VS **then**

begin

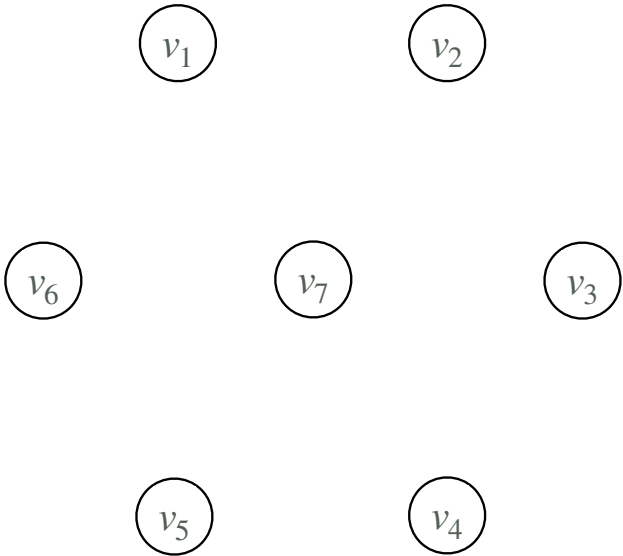
 9. replace W_1 and W_2 in VS by $W_1 \cup W_2$;
 10. add (v, w) to T

end

end
- end**

Minimum Spanning Tree

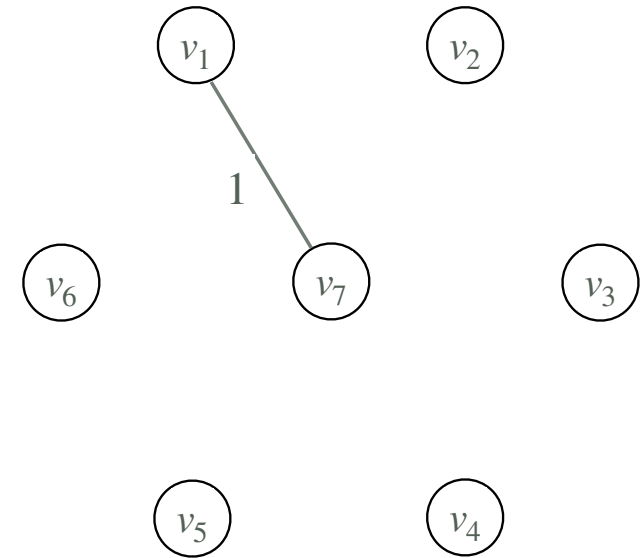
Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
(v_3, v_7)	9	(v_5, v_7)	25
(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36



$$\| \text{VS} \| = 7$$

Minimum Spanning Tree

Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
(v_3, v_7)	9	(v_5, v_7)	25
(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36

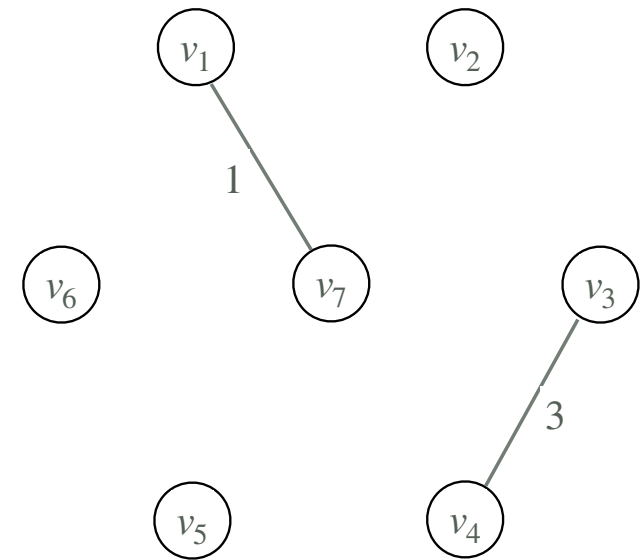


♦ Add (v_3, v_4) ; $VS = \{\{v_1, v_7\}, \{v_2\}, \{v_3\}, \{v_4\}, \{v_5\}, \{v_6\}\}$

♦ $\| VS \| = 6$

Minimum Spanning Tree

Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
(v_3, v_7)	9	(v_5, v_7)	25
(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36

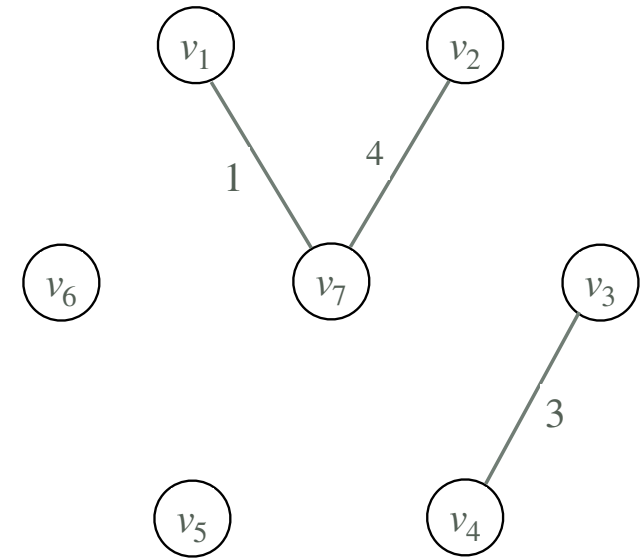


♦ Add (v_3, v_4) ; $VS = \{\{v_1, v_7\}, \{v_2\}, \{v_3, v_4\}, \{v_5\}, \{v_6\}\}$

♦ $\| VS \| = 5$

Minimum Spanning Tree

Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
(v_3, v_7)	9	(v_5, v_7)	25
(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36

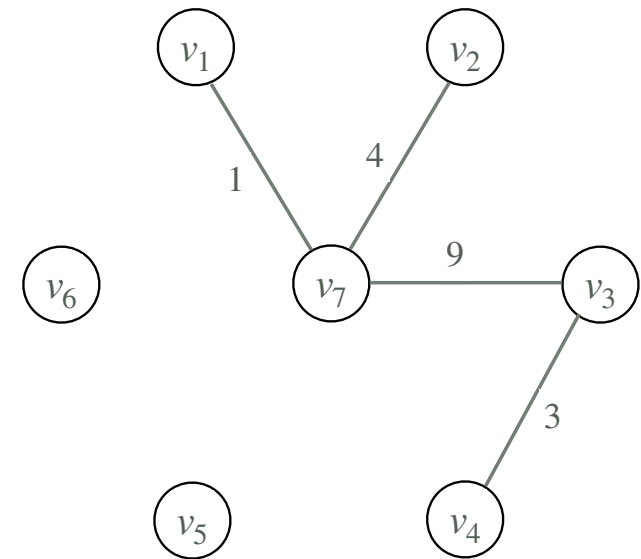


♦ Add (v_2, v_7) ; $VS = \{\{v_1, v_2, v_7\}, \{v_3, v_4\}, \{v_5\}, \{v_6\}\}$

♦ $\| VS \| = 4$

Minimum Spanning Tree

Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
(v_3, v_7)	9	(v_5, v_7)	25
(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36

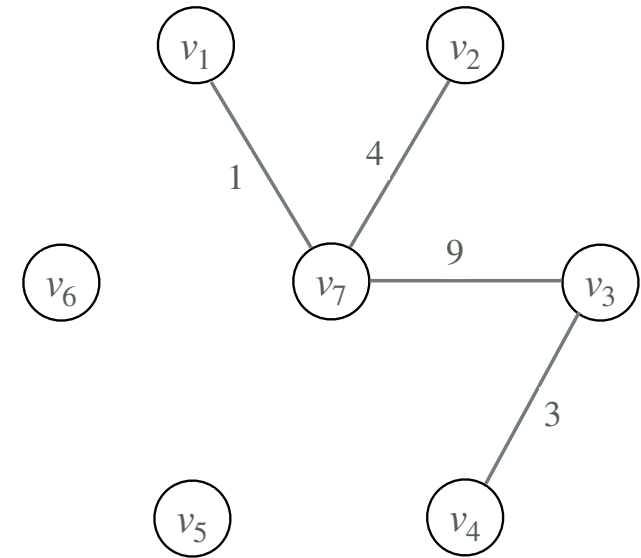


♦ Add (v_3, v_7) ; $VS = \{\{v_1, v_2, v_3, v_4, v_7\}, \{v_5\}, \{v_6\}\}$

♦ $\| VS \| = 3$

Minimum Spanning Tree

Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
(v_3, v_7)	9	(v_5, v_7)	25
(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36

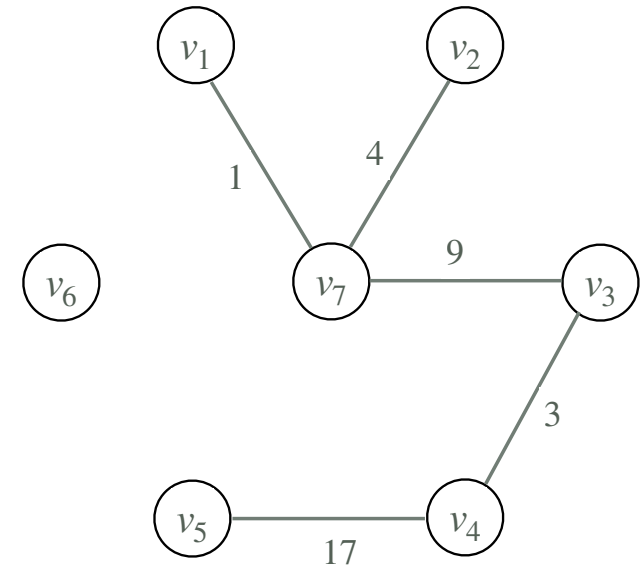


◆ Reject (v_2, v_3)

◆ Reject (v_4, v_7)

Minimum Spanning Tree

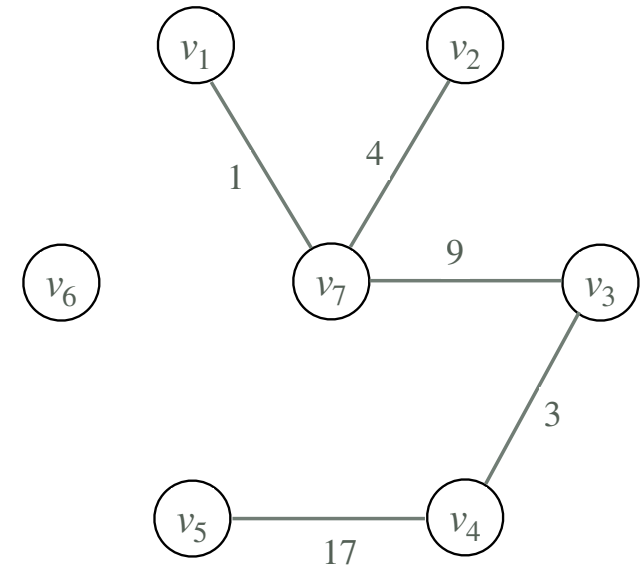
Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
(v_3, v_7)	9	(v_5, v_7)	25
(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36



- ♦ Add (v_4, v_5) ; $VS = \{\{v_1, v_2, v_3, v_4, v_5, v_7\}, \{v_6\}\}$
- ♦ $\| VS \| = 2$

Minimum Spanning Tree

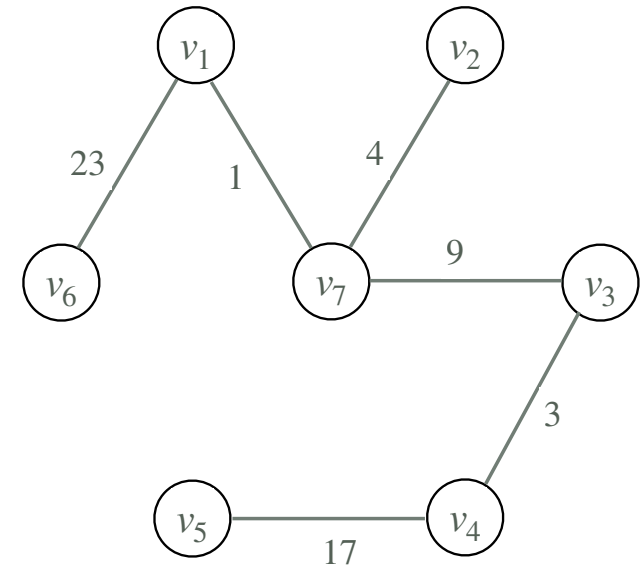
Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
(v_3, v_7)	9	(v_5, v_7)	25
(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36



◆ Reject (v_1, v_2)

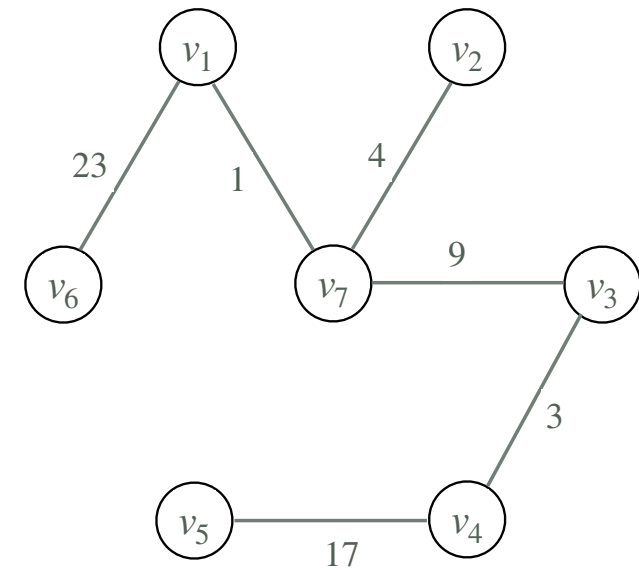
Minimum Spanning Tree

Edge	Cost	Edge	Cost
(v_1, v_7)	1	(v_4, v_5)	17
(v_3, v_4)	3	(v_1, v_2)	20
(v_2, v_7)	4	(v_1, v_6)	23
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(v_2, v_3)	15	(v_5, v_6)	28
(v_4, v_7)	16	(v_6, v_7)	36



- ♦ Add (v_1, v_6) ; $VS = \{\{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}\}$
- ♦ $\| VS \| = 1$

Minimum Spanning Tree



♦ Time complexity?

♦ $O(|E| \log |E|)$

begin

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end

end

end